



Year 7

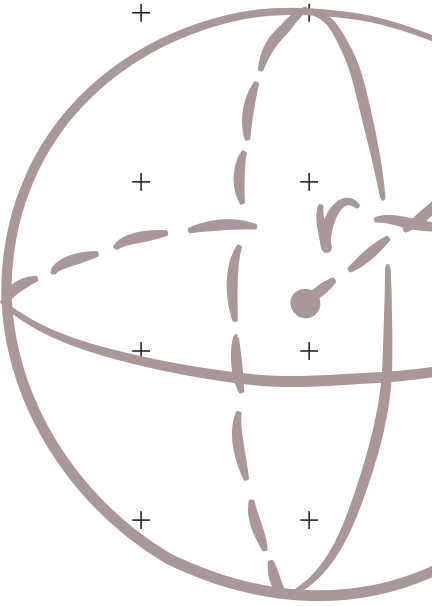
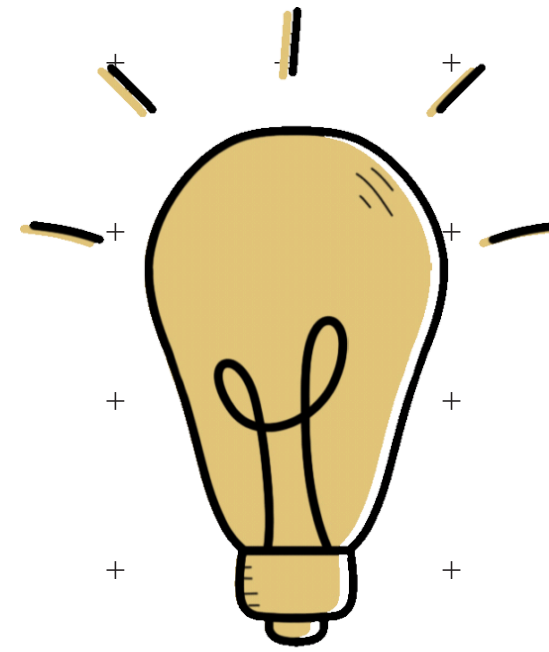
All about Measurement



LET'S START WITH SOME

MATHS

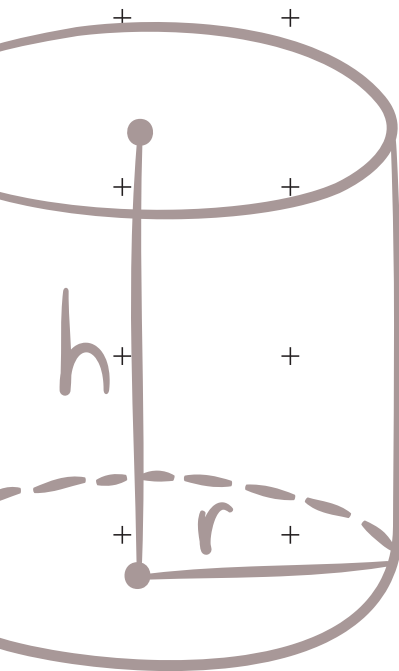
BRAIN TEASERS



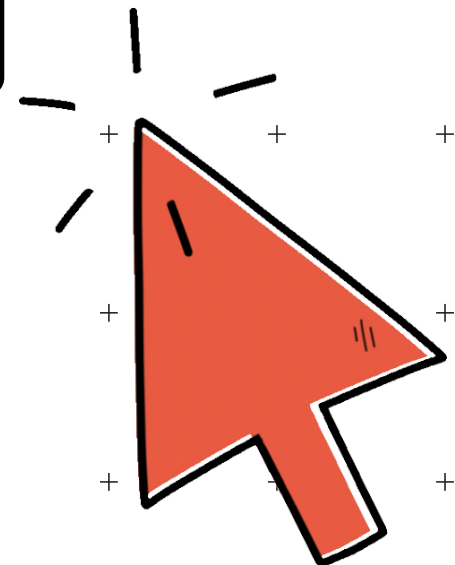
$$V = \frac{4}{3} \pi r^3$$

Your paragraph text

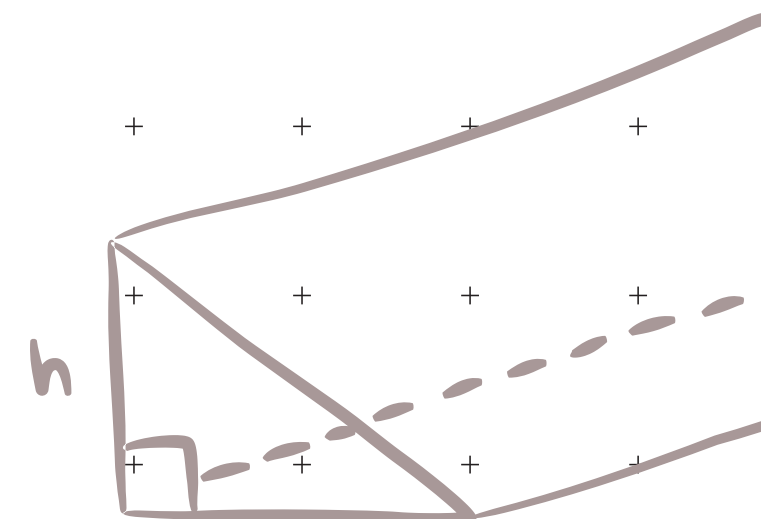
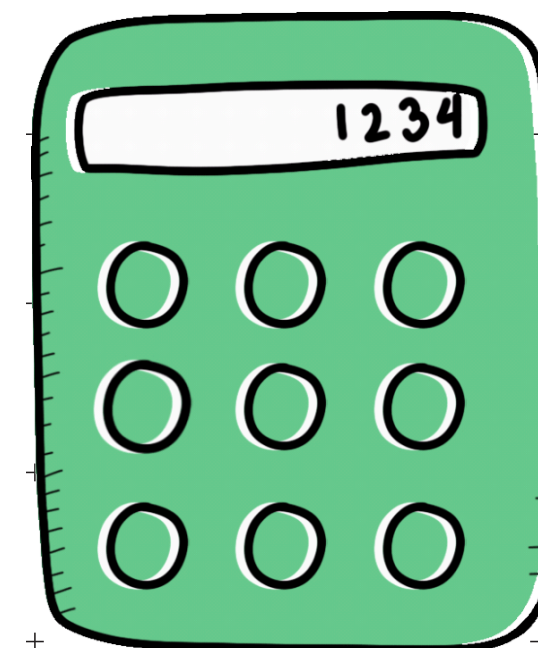
Start Now



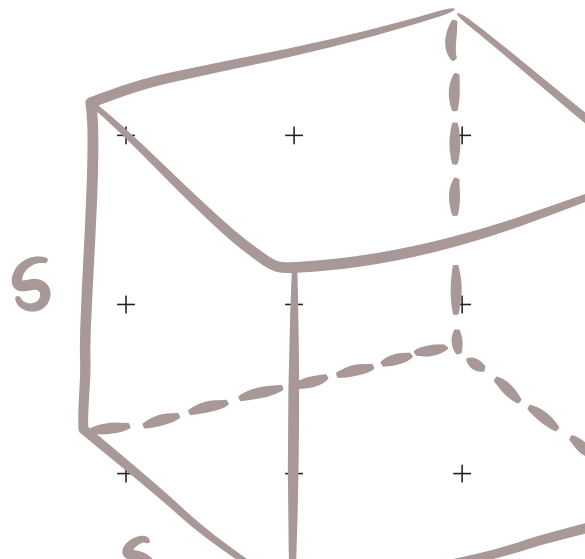
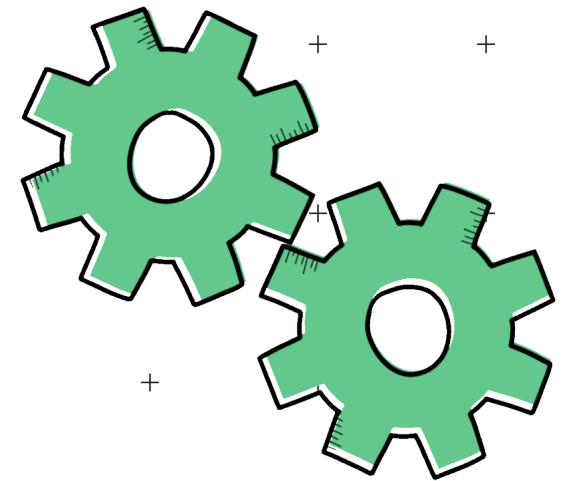
$$V = \pi r^2 h$$



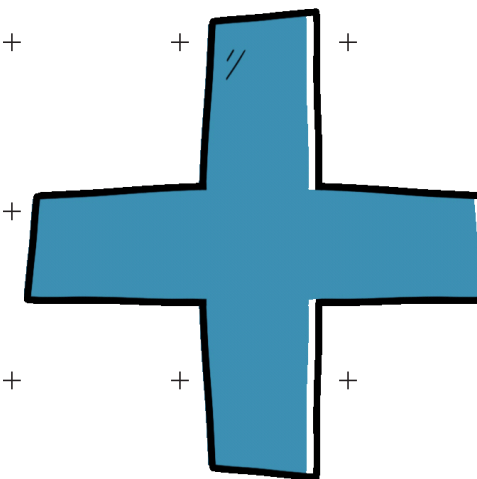
Q1: Ben is 6 years old.
His sister is twice
as old as him. When
Ben is 50, how old
will his sister be?



Q2: Which weighs more:
10 kg of brick or
10 kg of feathers?



Q3: I am a three digit number. My second digit is three times larger than my third digit. My digits sum to five. What number am I?

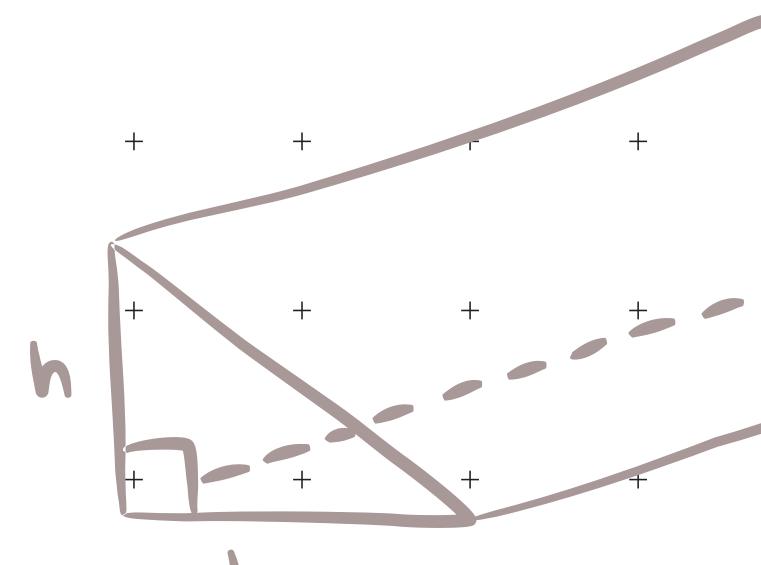
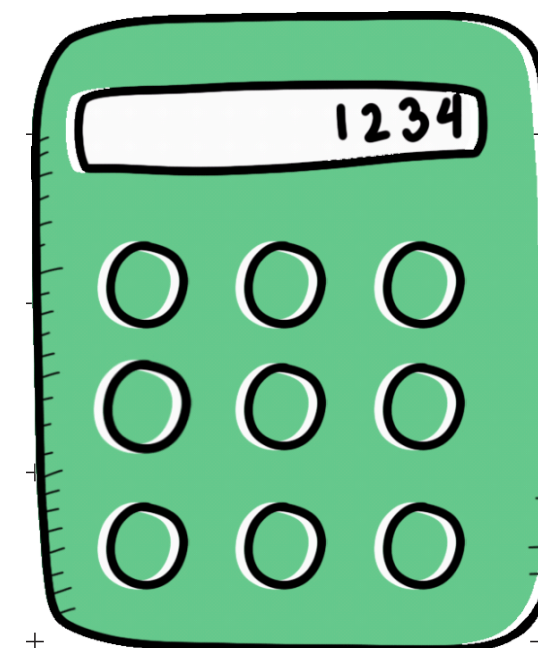


$$x = \frac{-b \pm \sqrt{b^2 - 2a}}{2a}$$

A:

Ben is 6 years old.
His sister is twice
as old as him. When
Ben is 50, how old
will his sister be?

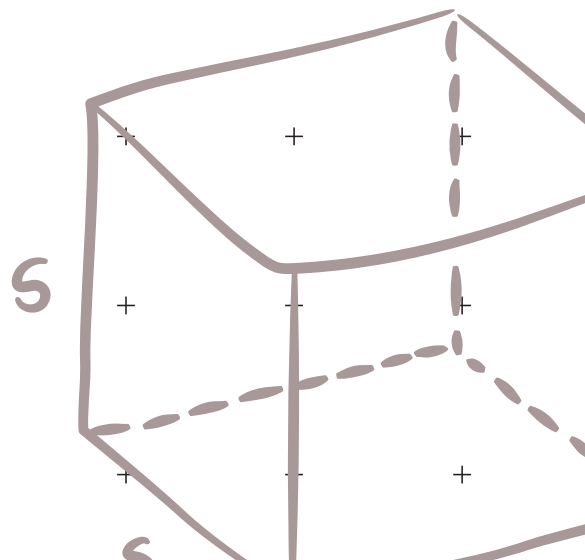
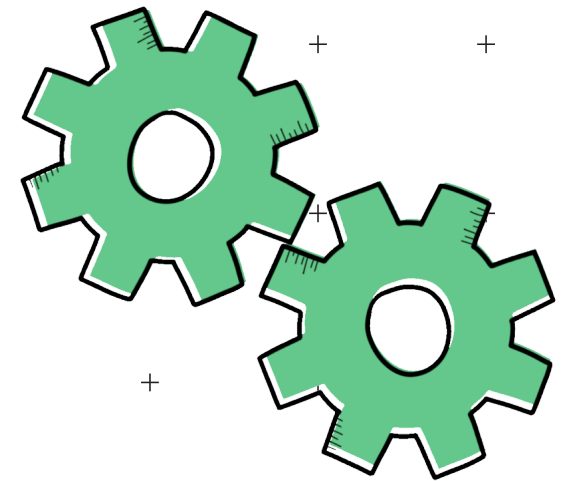
56 YEARS OLD



A:

Which weighs more:
10 kg of brick or
10 kg of feathers?

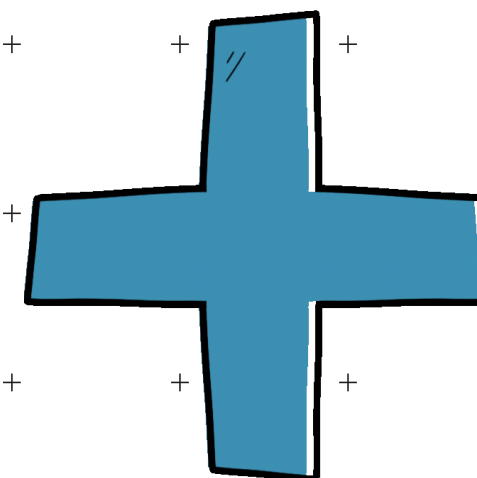
**BOTH WEIGH
THE SAME**



A:

I am a three digit number. My second digit is three times larger than my third digit. My digits sum to five. What number am I?

131



$$x = \frac{-b \pm \sqrt{b^2 - 2a}}{2a}$$

Learning Outcomes

Measure objects using correct units.

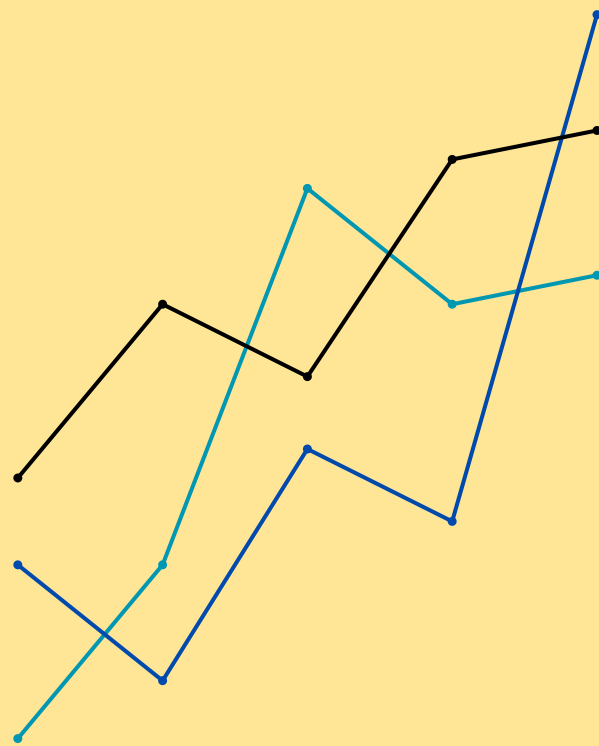
Find perimeter, area, and volume.

Convert units and solve problems.

IMPORTANCE OF PRECISION IN MEASUREMENT



ACCURATE DATA



Precise measurements lead to more accurate data, which is crucial for making the correct decisions, do experiments properly, check results, and make sure things like products or projects turn out just right.

COMPARABILITY



When measurements are precise, different researchers or laboratories can obtain similar results, allowing for meaningful comparisons and collaborations across different studies and institutions.

MINIMIZING ERRORS

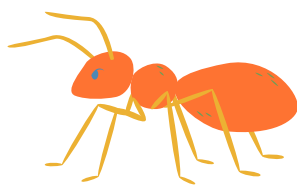


Measuring very carefully helps us avoid mistakes that can happen because of tools, people, or the surroundings. This way, our results are more reliable and trustworthy.

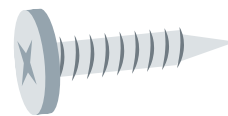
Units of Length

mm

Millimetres are used to measure very small objects



2mm



15 mm

cm

Centimetres are used to measure small to medium sized objects



15cm

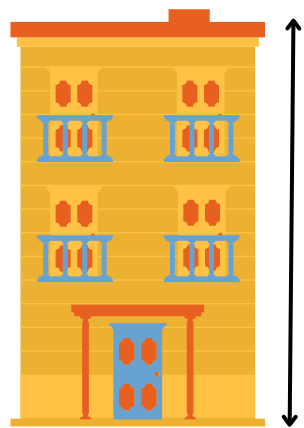


80 cm

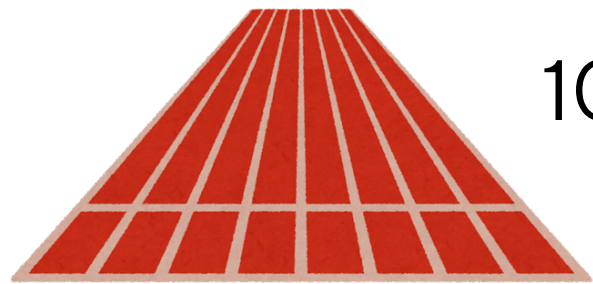
Units of Length

m

Metres are used to measure large objects and short distances



10m



100 m

km

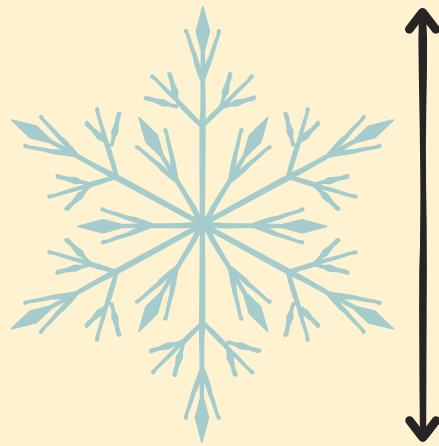
Kilometres are used to measure long distances



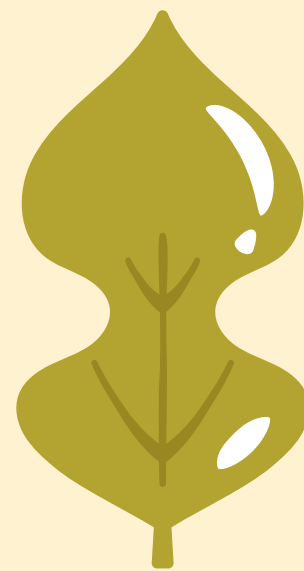
12 000 km

Which Unit?

Which unit of length would you use to measure the size of the following objects?



mm



cm



m

How to estimate the size of different objects:

1. Decide on the unit

Choose the appropriate unit of length: use centimetres for shorter objects and metres for longer ones.

2. Use a reference object

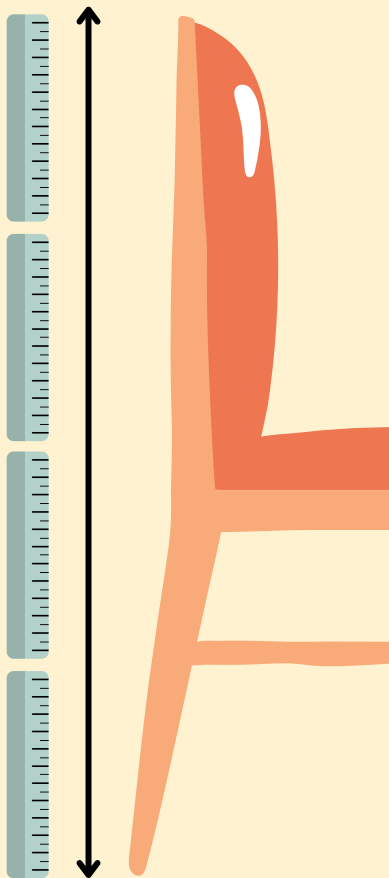
Pick something you know the length of, such as a ruler or the size of your hand.

3. Count how many fit

Roughly count how many times the reference object will fit along the length of the item.

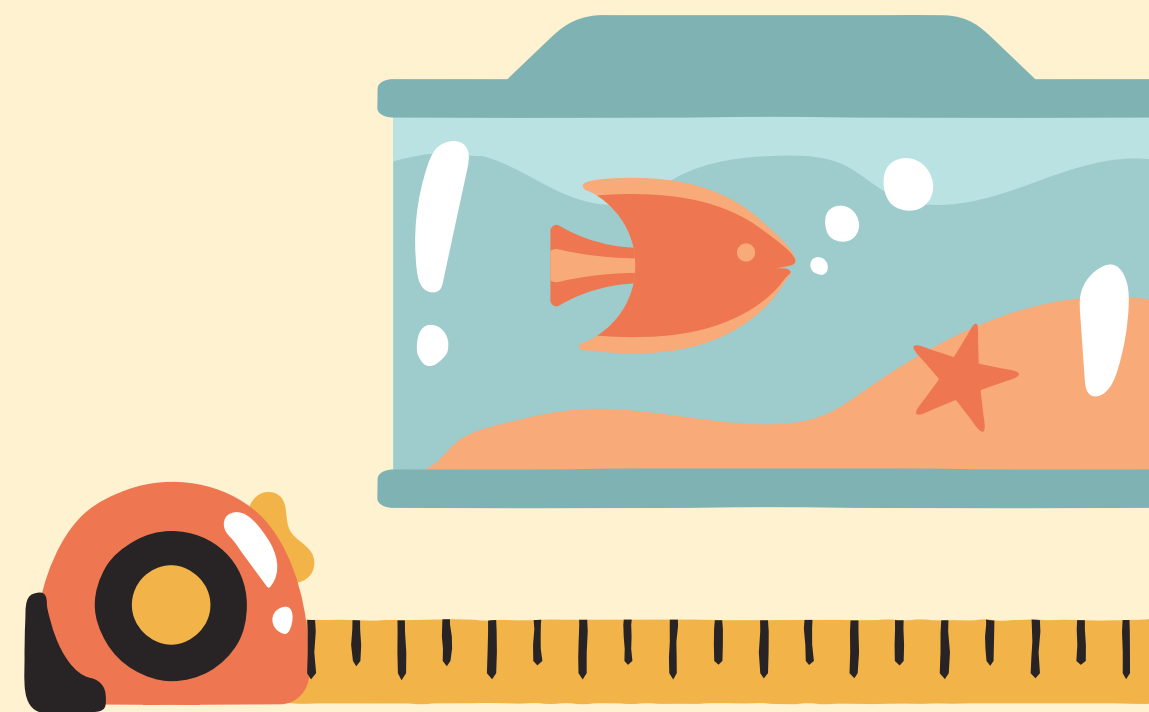
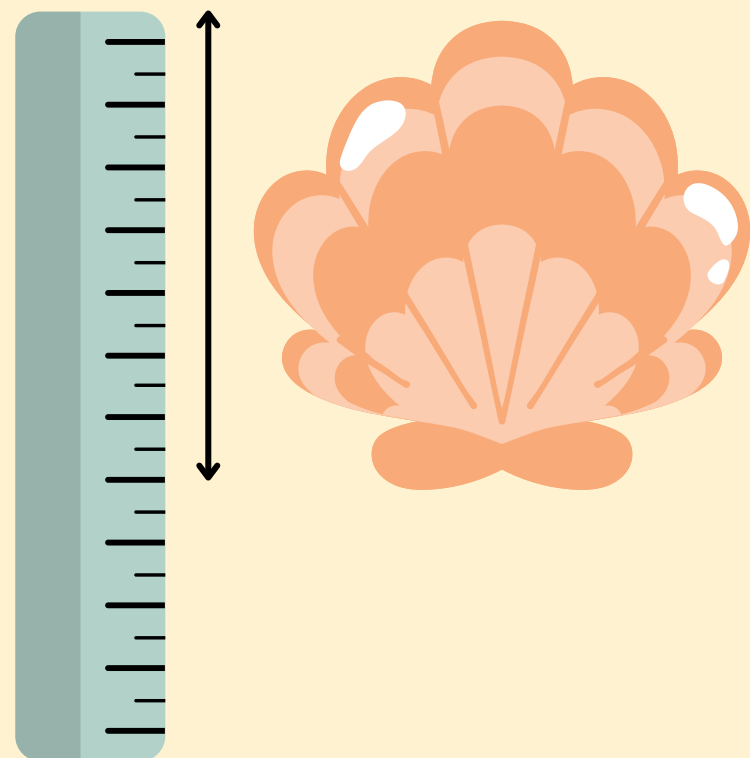
4. Multiply and estimate

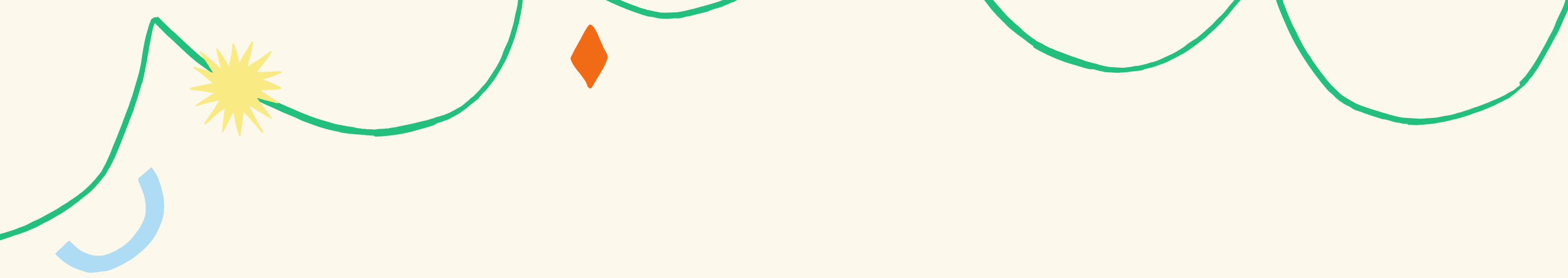
Multiply the reference object's length by how many times it fits along the item's length.



How to measure length using a ruler:

1. Place the ruler next to the object.
2. Put the end of the object in line with the "0" on the ruler.
3. Keep the ruler straight and read the measurement at the other end of the object

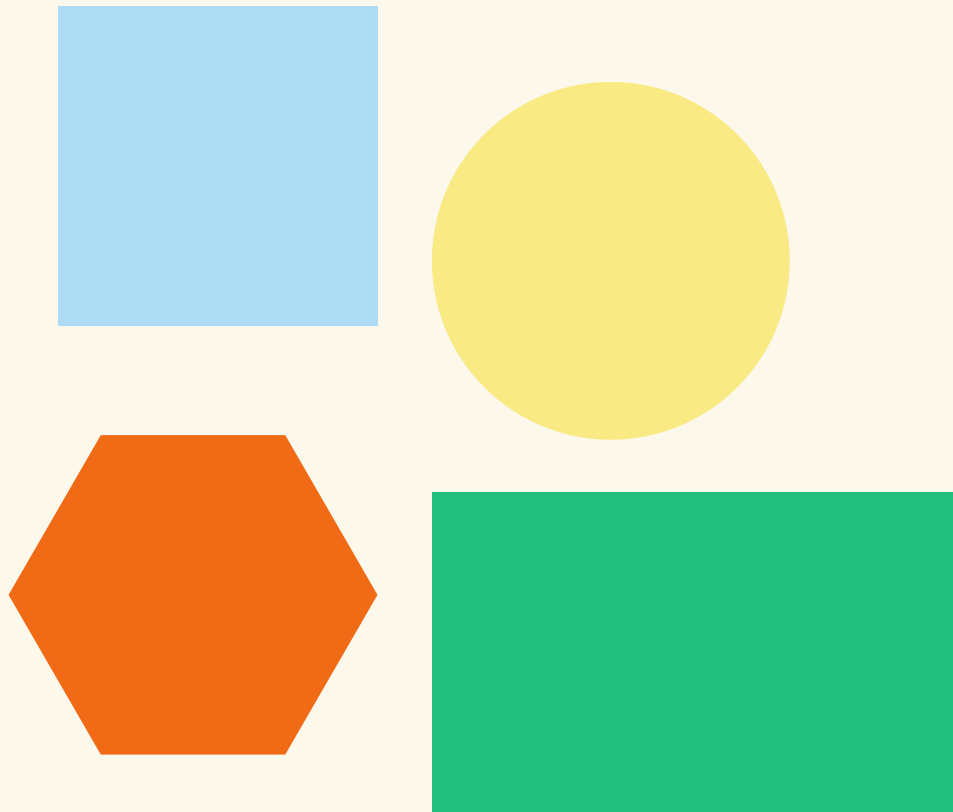




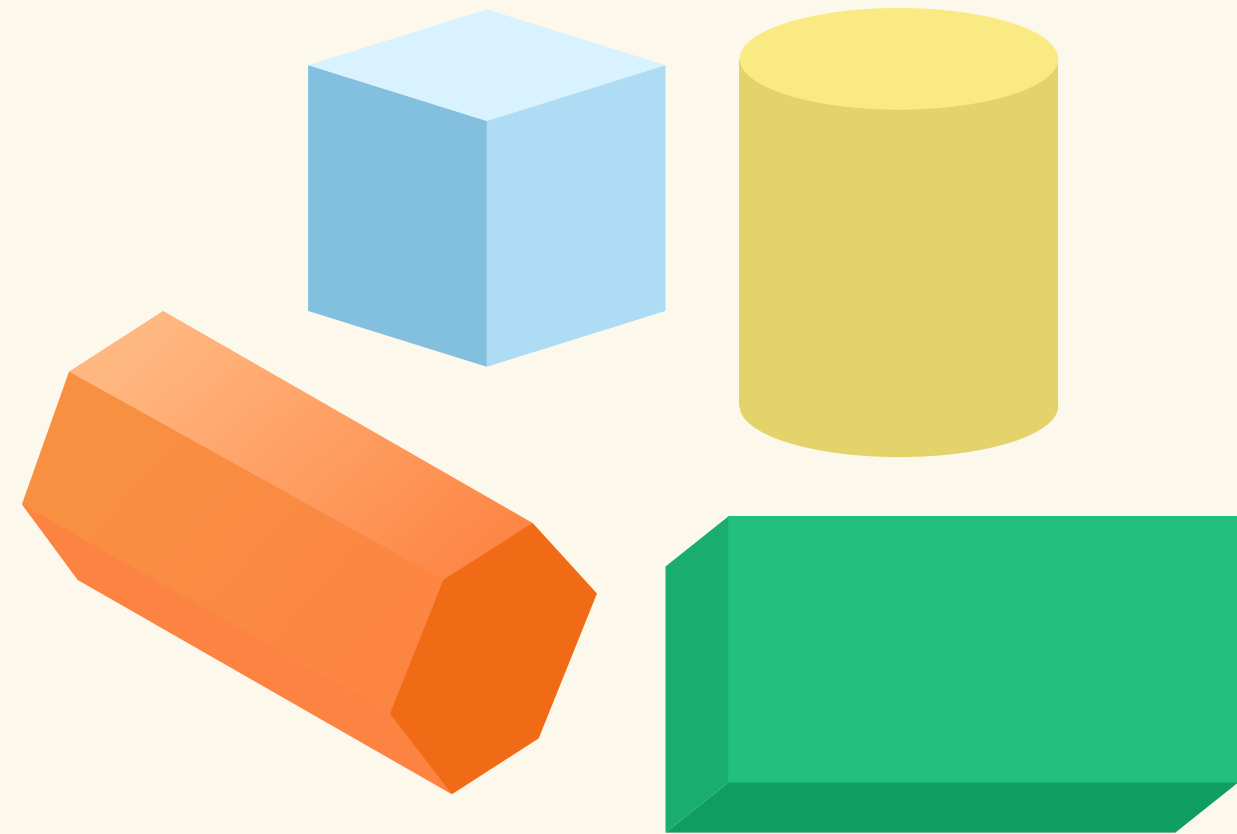
Surface Area and Volume of Solid Figures



What is the difference between the two sets of figure below?

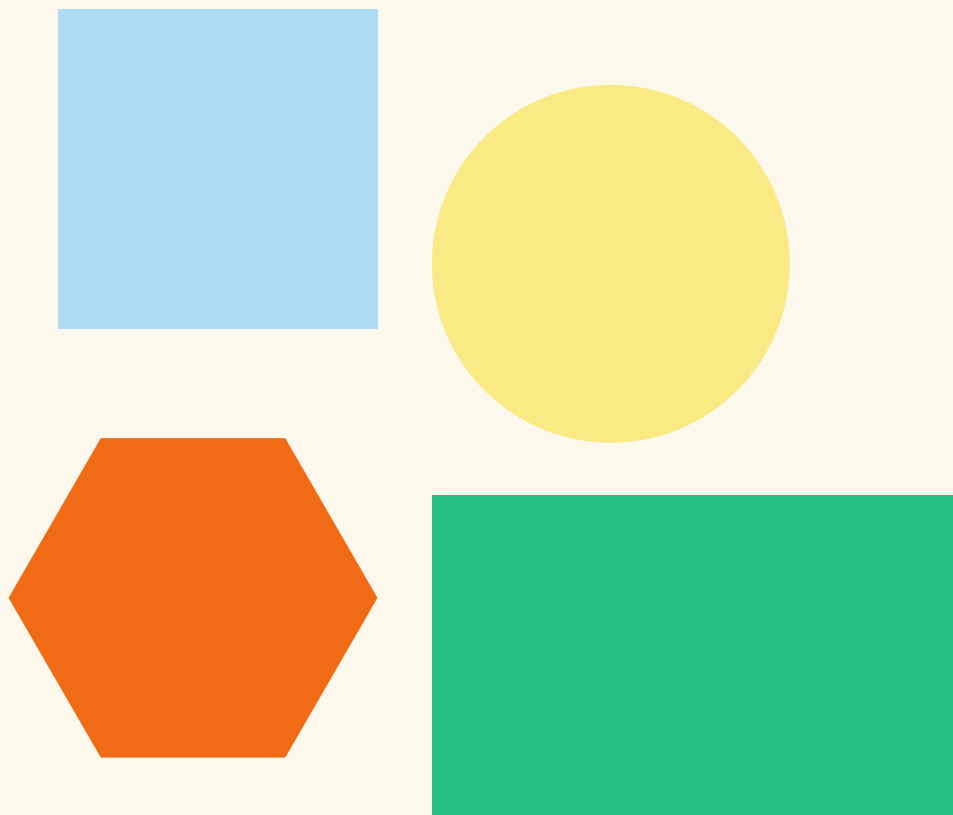


Set A



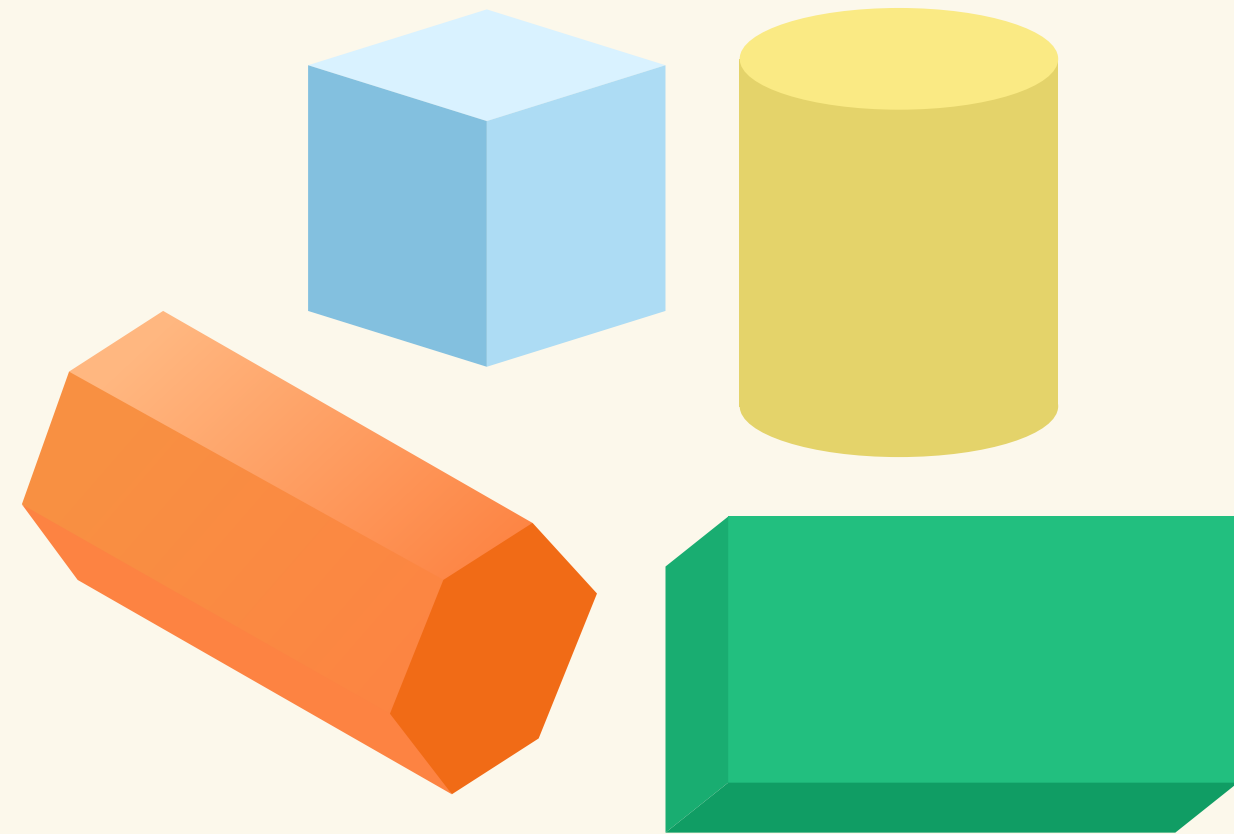
Set B

What is the difference between the two sets of figure below?



Set A

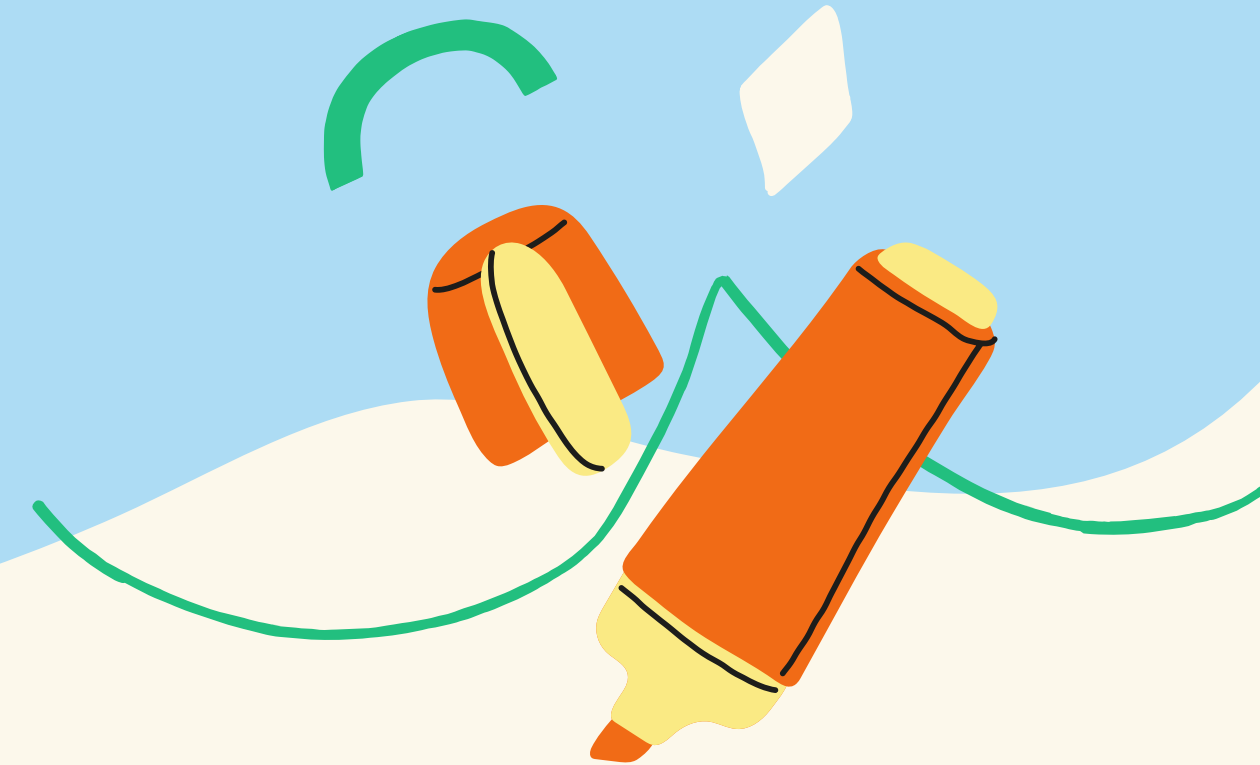
2D/ Plane Figures



Set B

3D/ Solid Figures

Let's define the following terms!



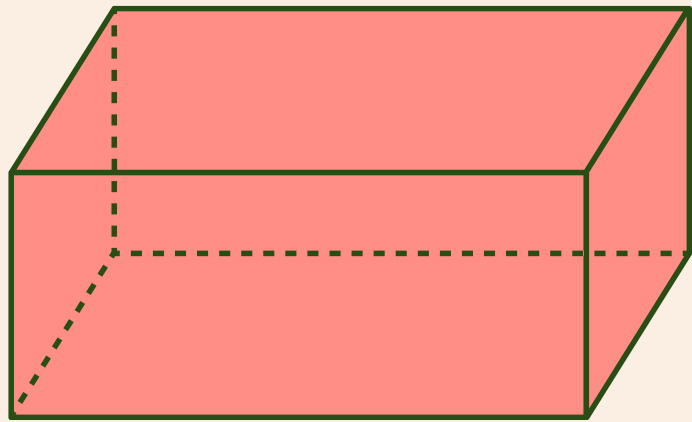
Surface Area

- ✱ It is the total area of all of the surfaces of a 3D figure.
- A cube has a surface area equal to the total area of six squares.

Use of Surface Area

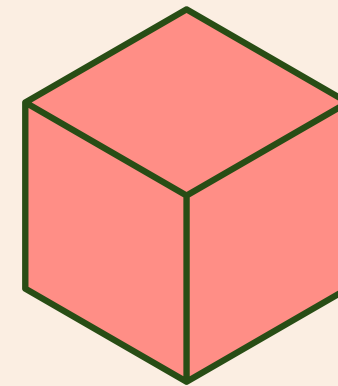
- ✱ It is the total area of all of the surfaces of a 3D figure.
- A cube has a surface area equal to the total area of six squares.

What is Volume?



Volume

the amount of space an object occupies

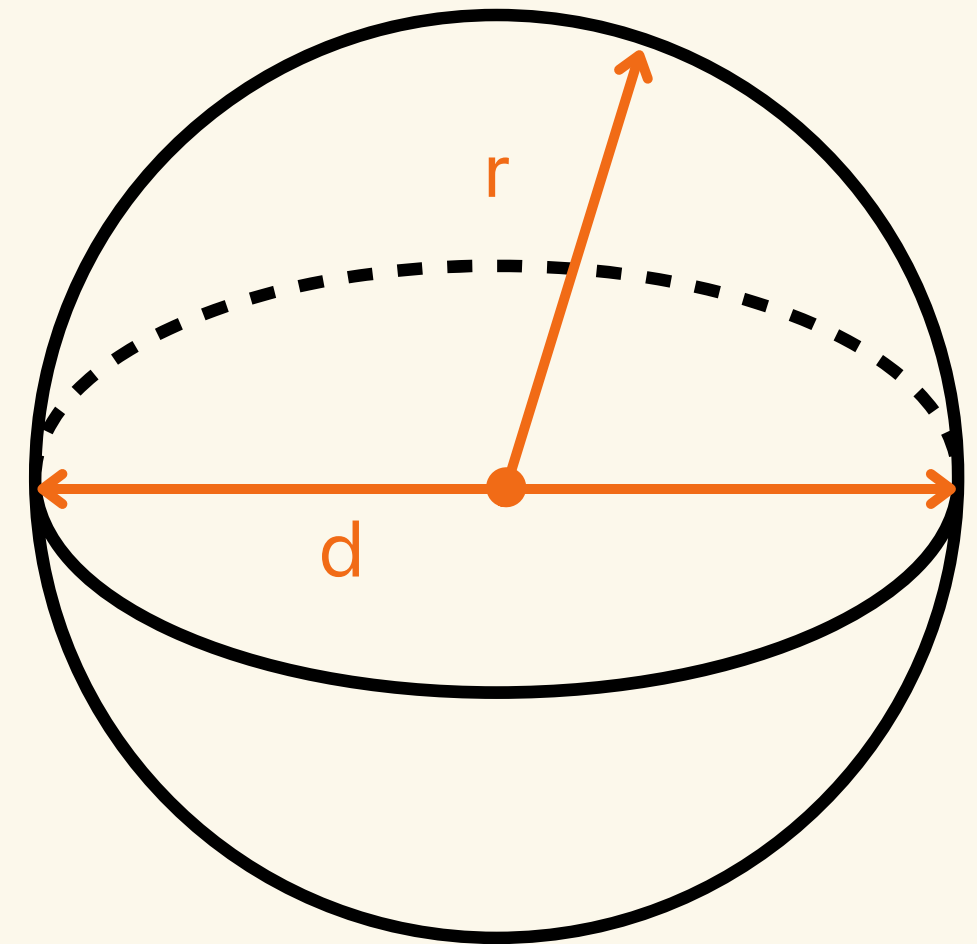


Unit Cube

a cube whose sides are equal to 1 unit long

Sphere

A sphere is a completely round figure. It has no faces, edges, or vertices.



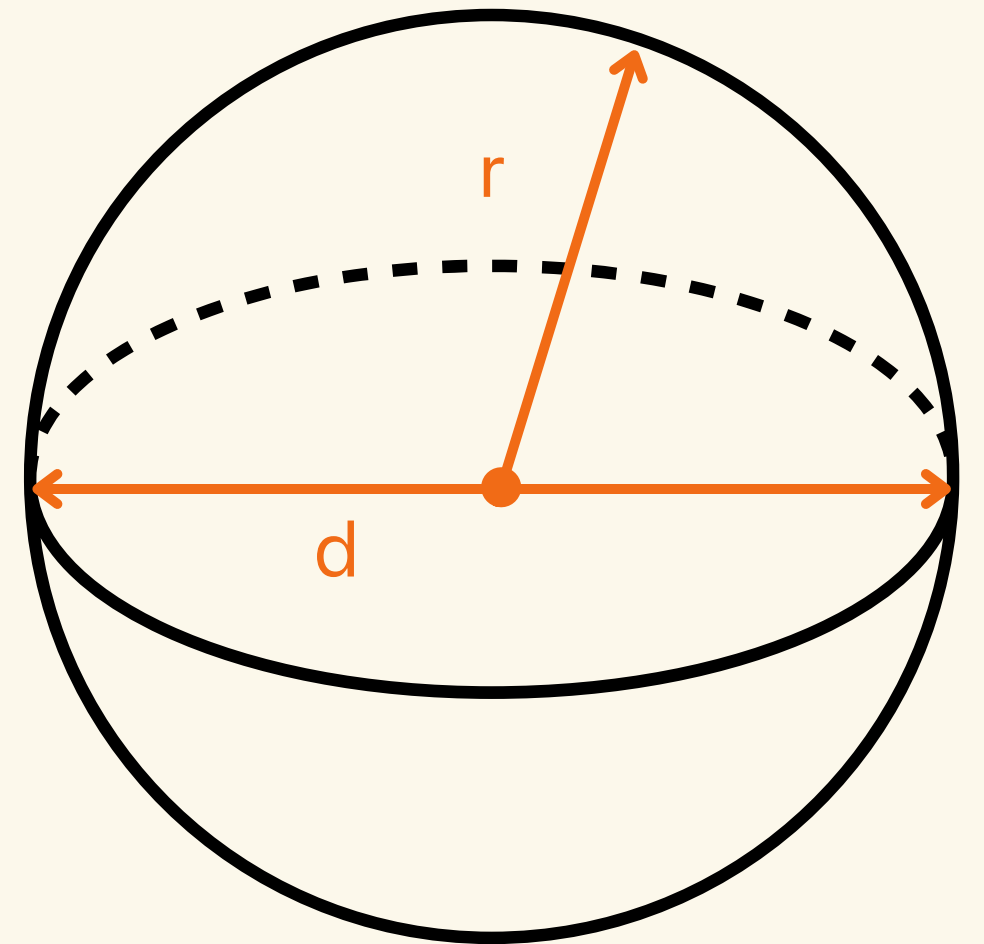
Sphere

Surface Area

$$A = 4\pi r^2$$

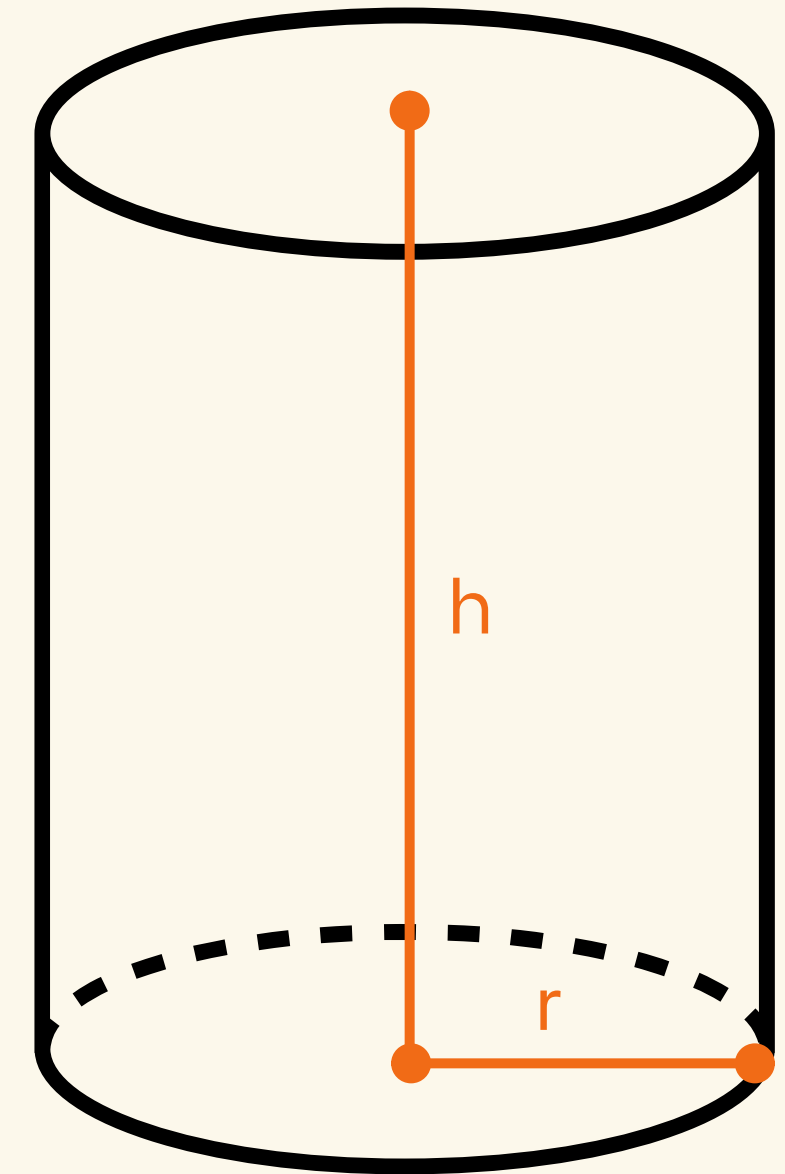
Volume

$$V = \frac{4}{3}\pi r^3$$



Cylinder

A cylinder has two circular faces but has no edges and vertices.



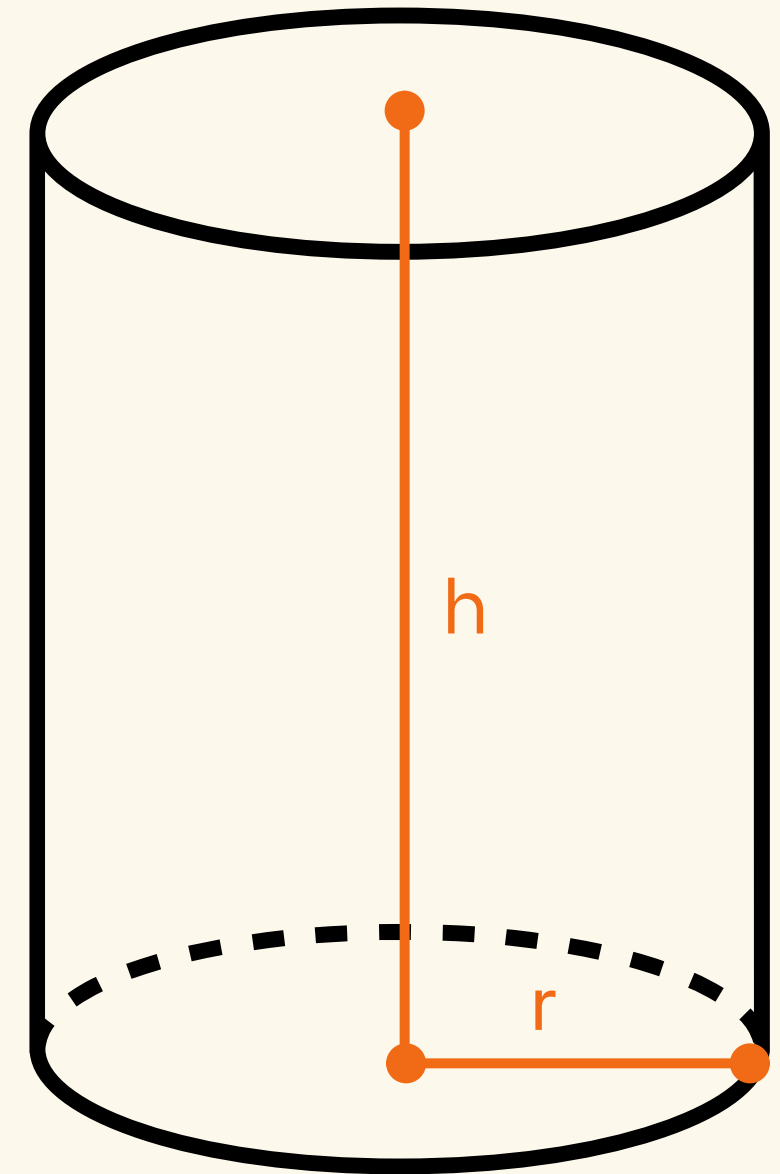
Cylinder

Surface Area

$$A = 2\pi rh + 2\pi r^2$$

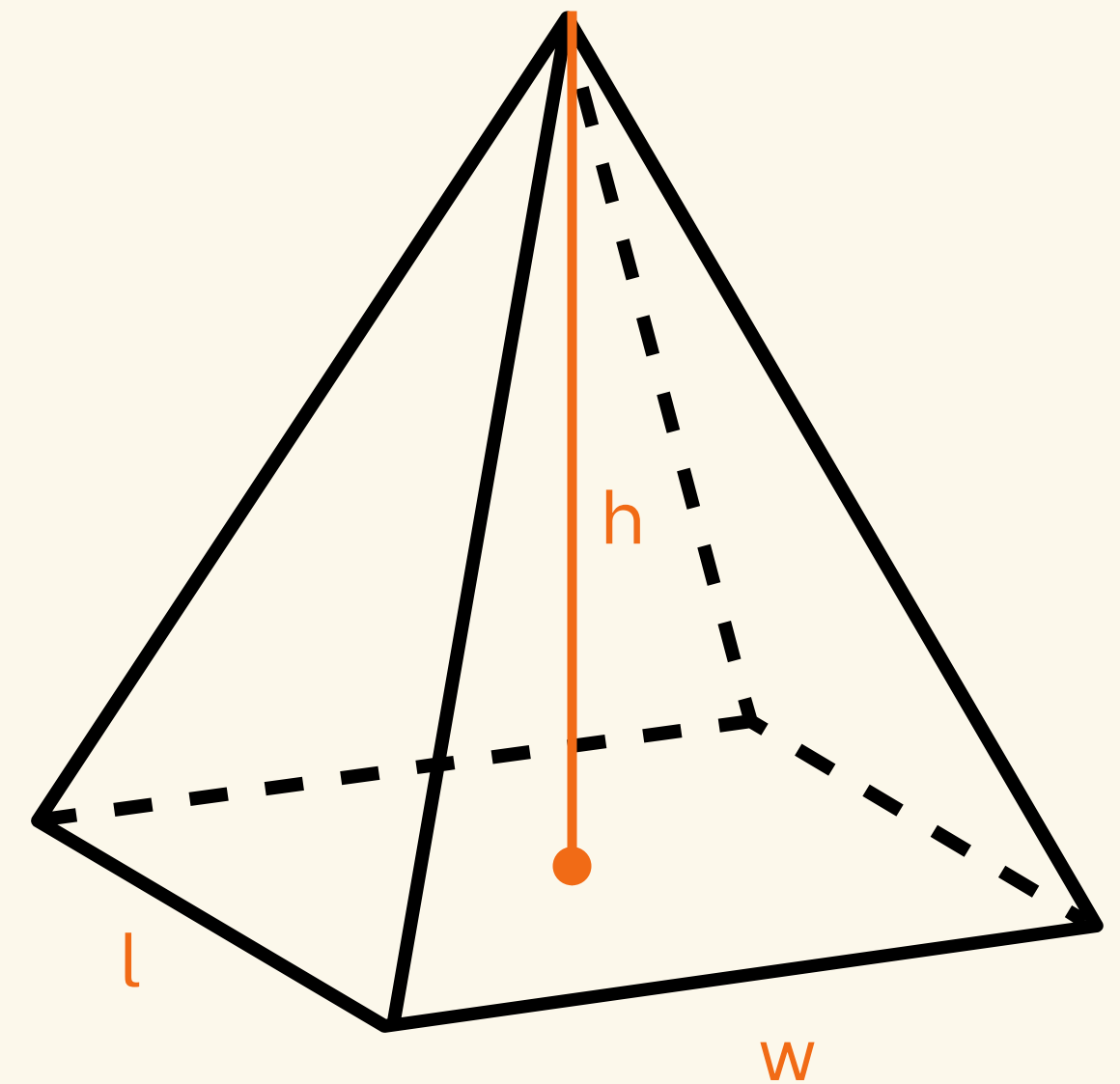
Volume

$$V = \pi r^2 h$$



Pyramid

A pyramid has one base and at least three triangular sides.



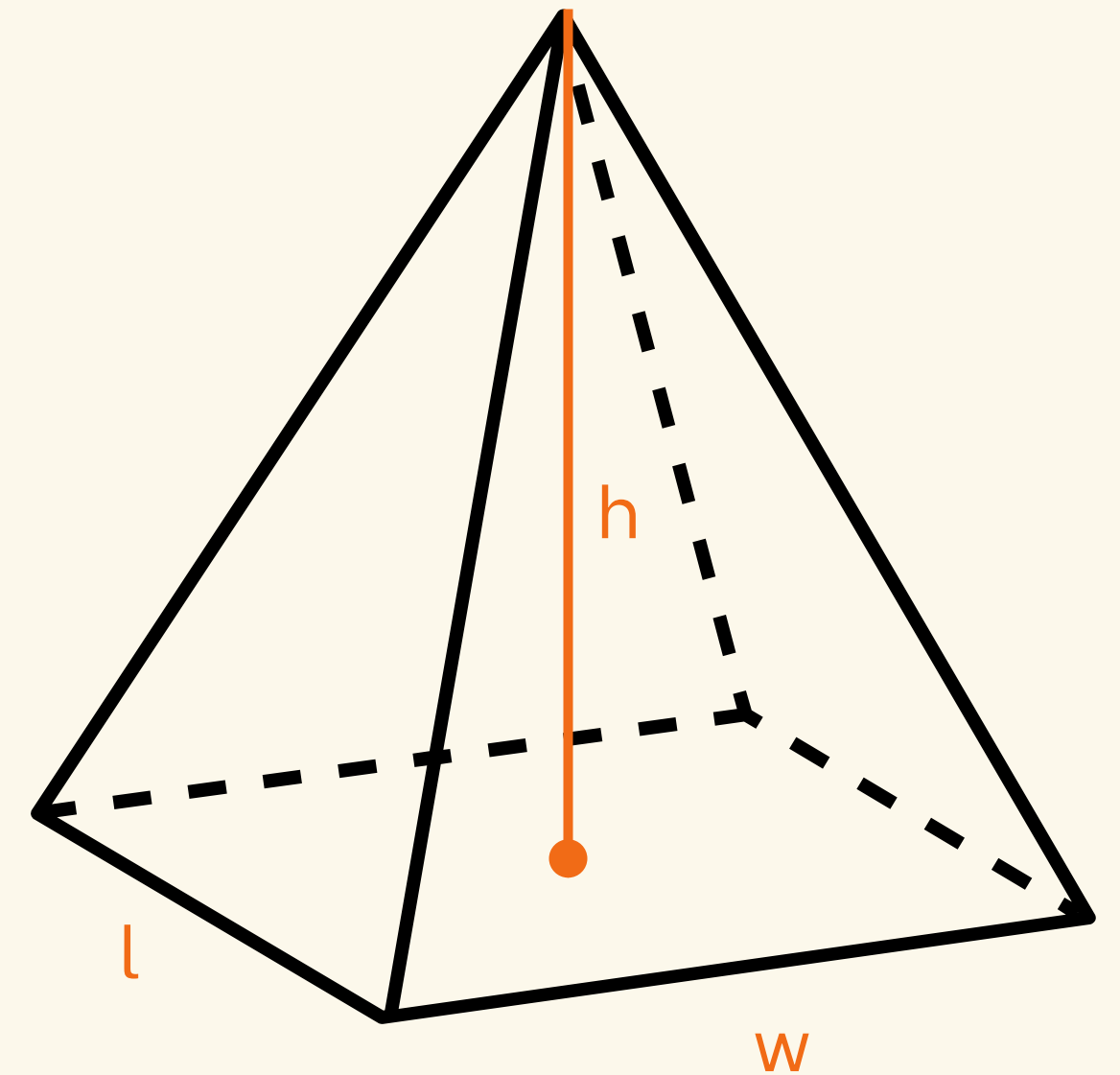
Pyramid

Surface Area

$$A = lw + l\sqrt{\left(\frac{w}{2}\right)^2 + h^2} + w\sqrt{\left(\frac{l}{2}\right)^2 + h^2}$$

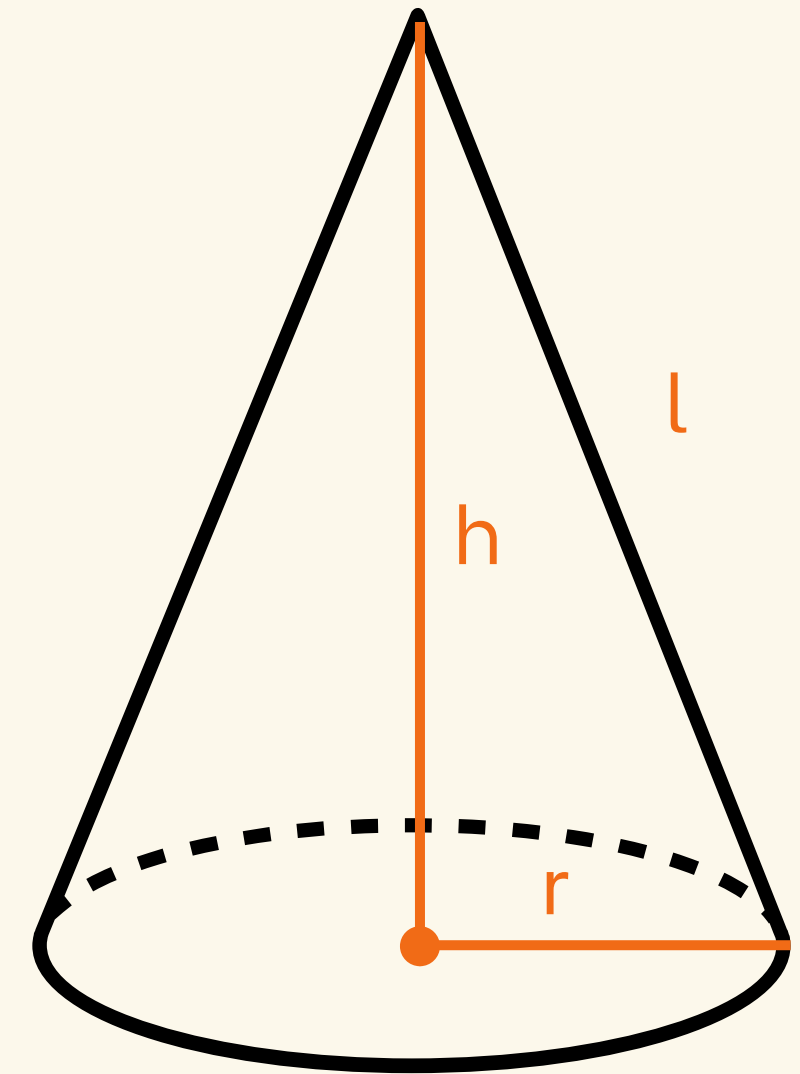
Volume

$$V = \frac{lwh}{3} \qquad A = \pi r \left(r + \sqrt{h^2 + r^2} \right)$$



Cone

A cone has one round face but has no edges or vertices.



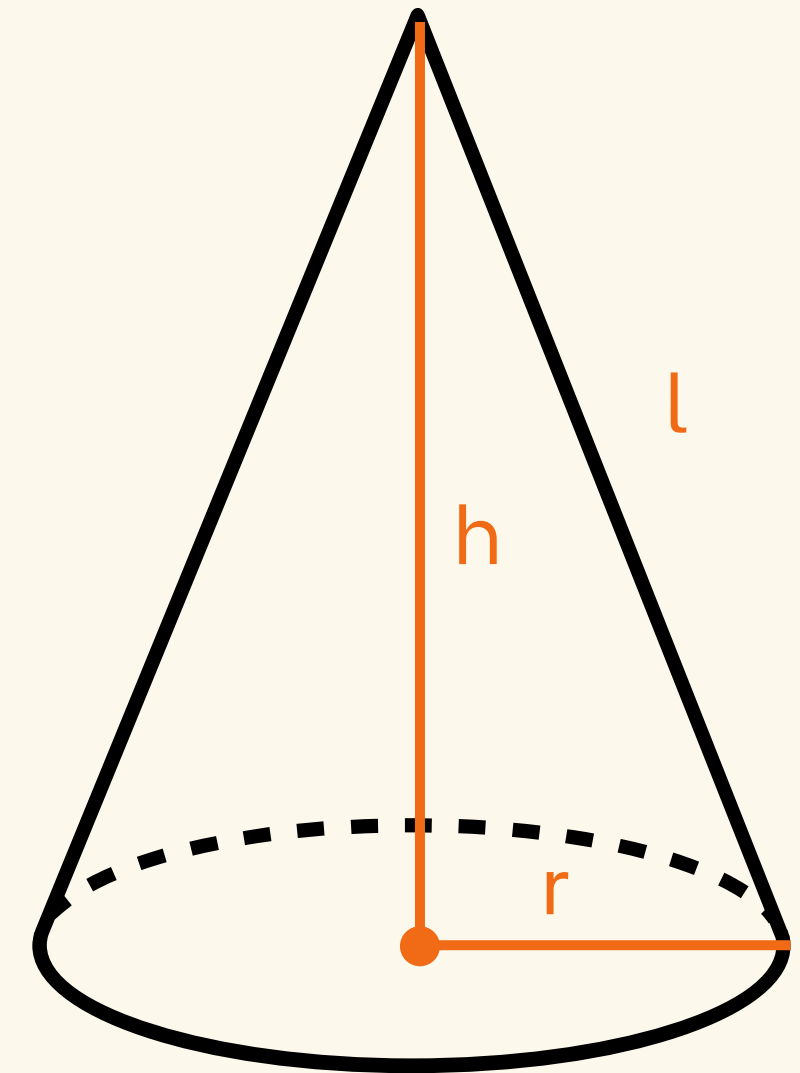
Cone

Surface Area

$$A = \pi r \left(r + \sqrt{h^2 + r^2} \right)$$

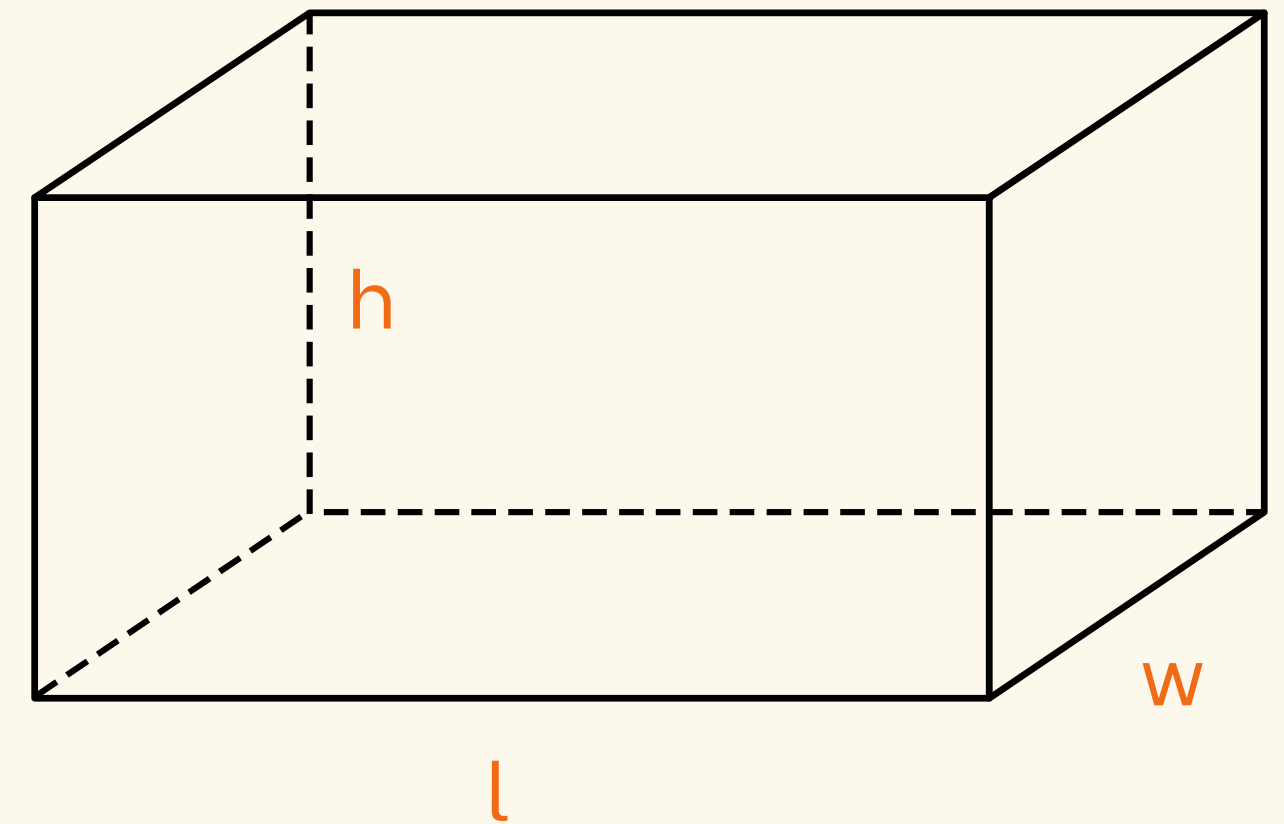
Volume

$$V = \pi r^2 \frac{h}{3}$$



Prism

A prism has two congruent parallel bases and some number of sides. A rectangular prism is one common example of prism.



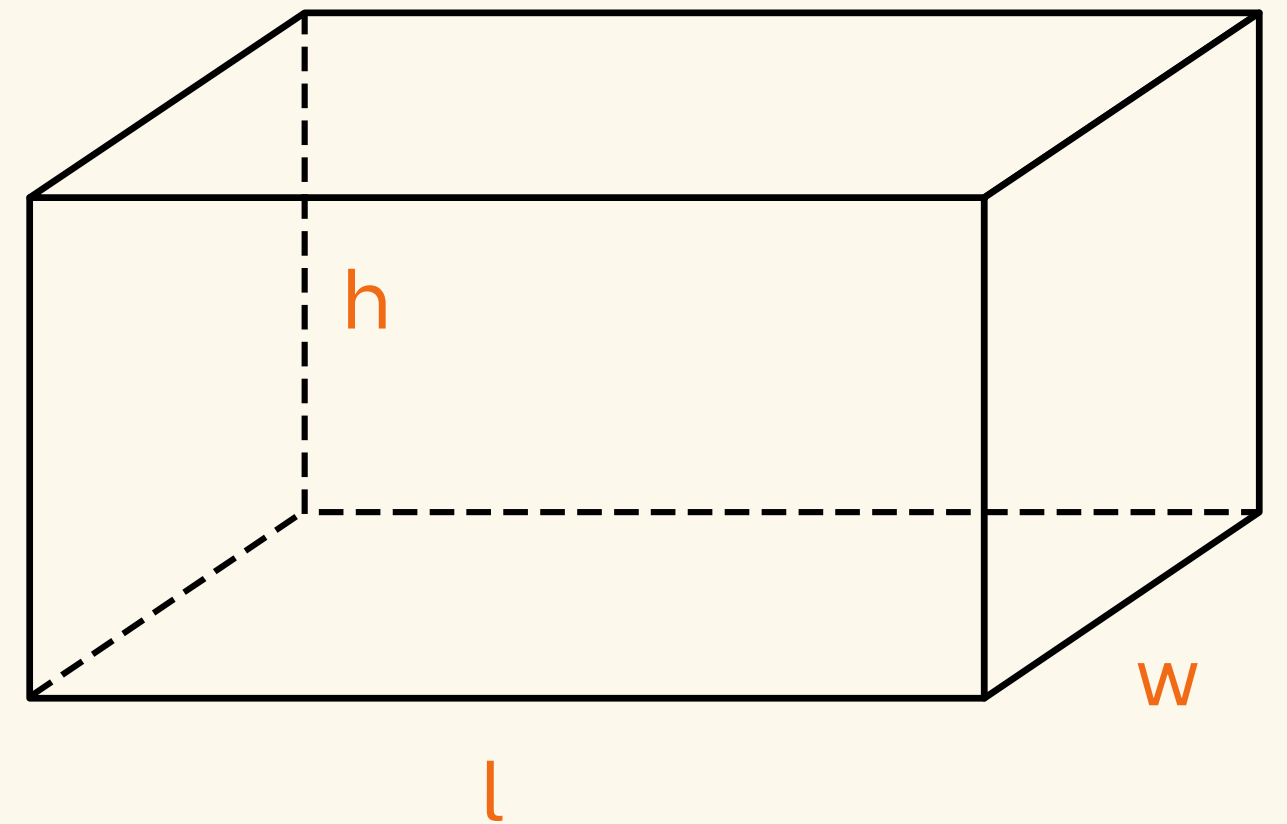
Prism

Surface Area

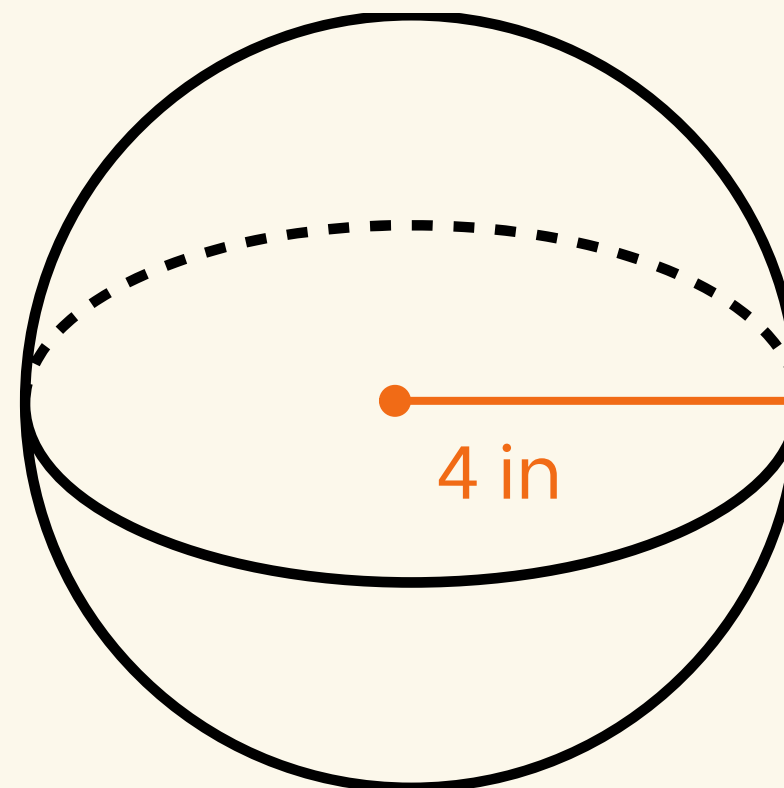
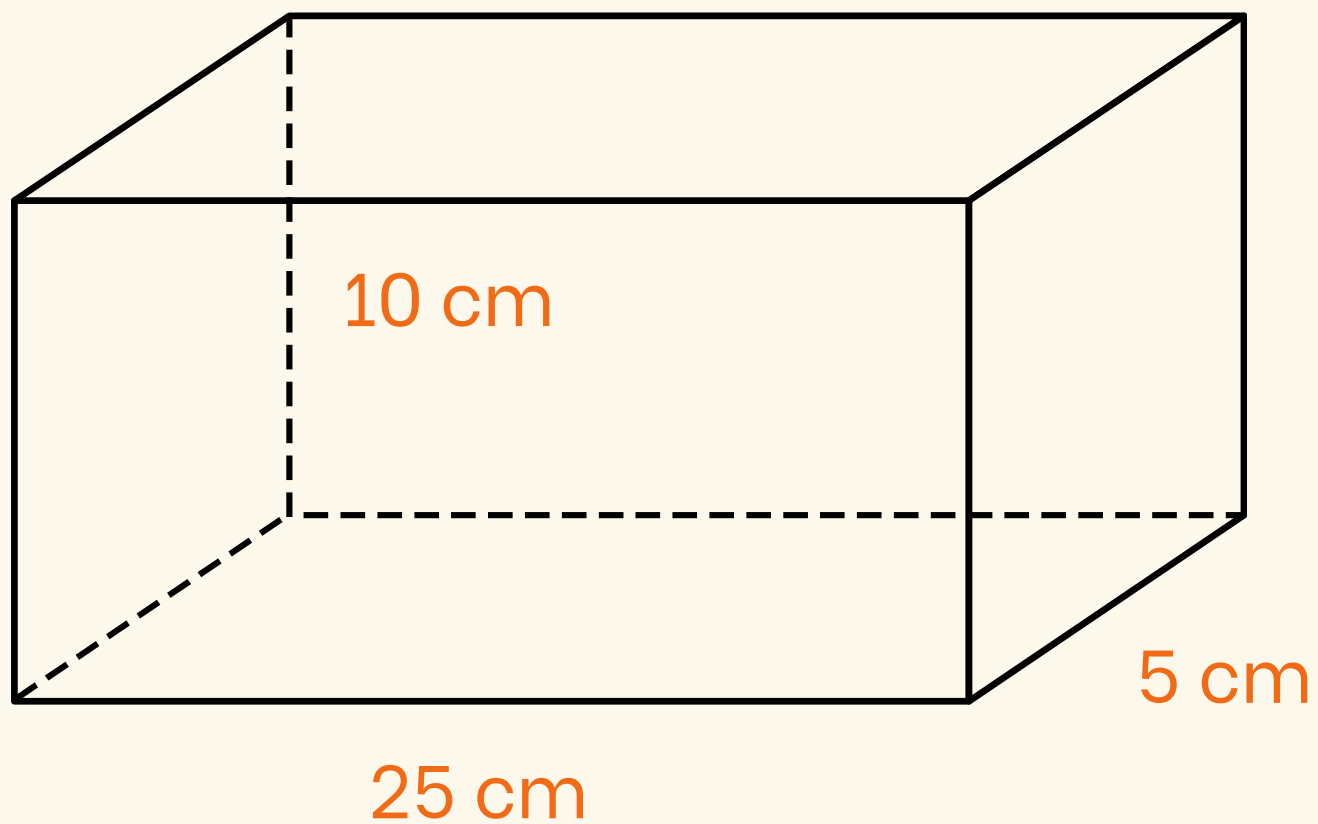
$$A = 2 (wl + hl + hw)$$

Volume

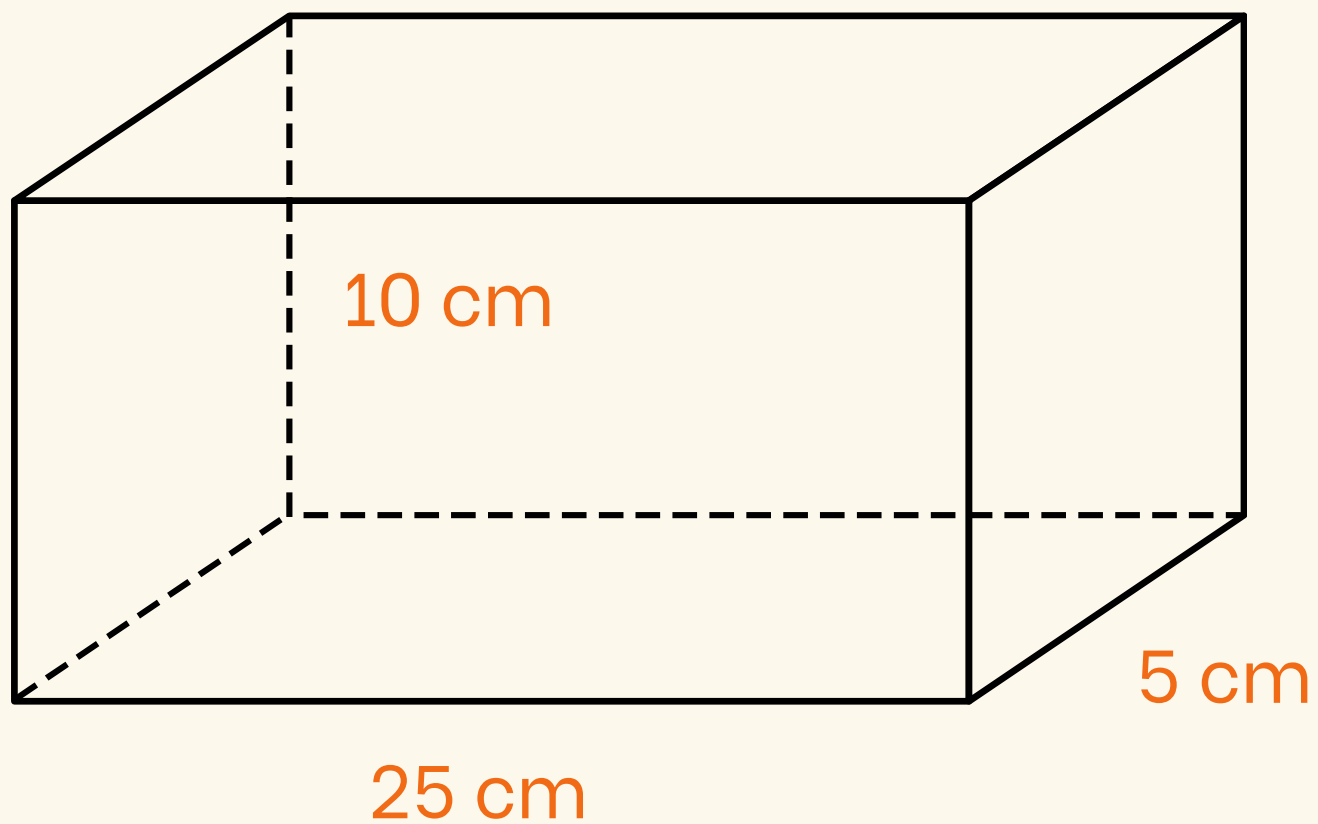
$$V = lwh$$



Classify each solid figure. Then, calculate the surface area of each figure.

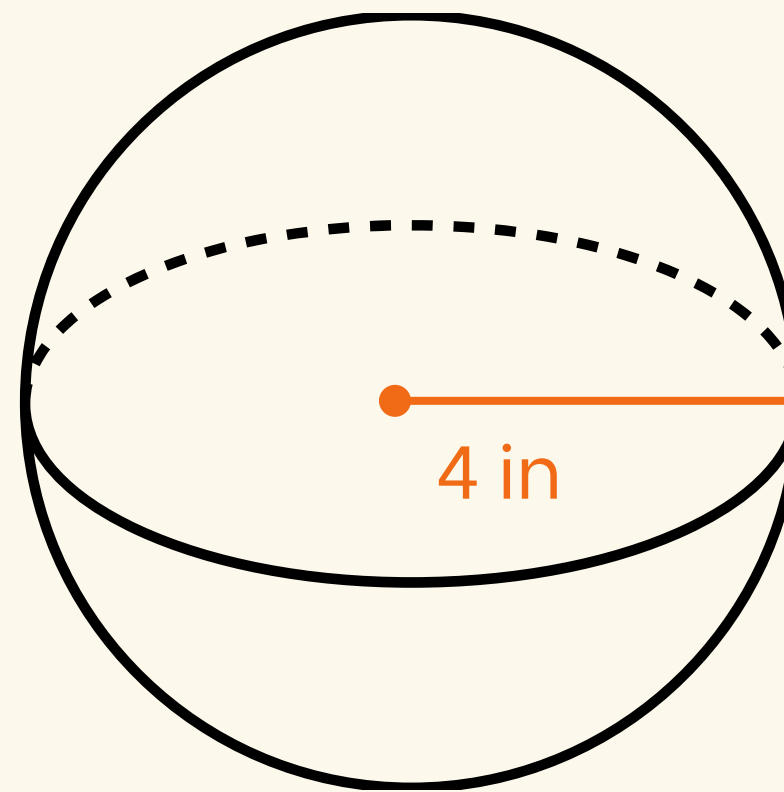


Classify each solid figure. Then, calculate the surface area of each figure.



Rectangular prism

$$A = 850\text{cm}^2$$

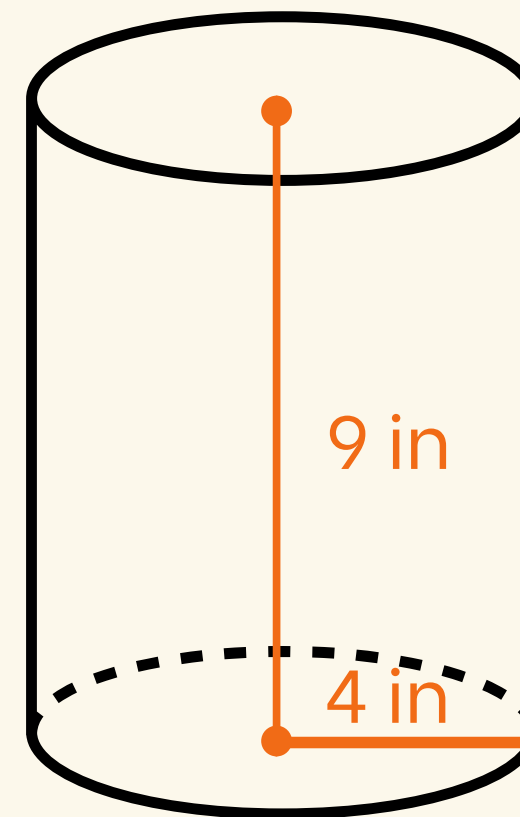
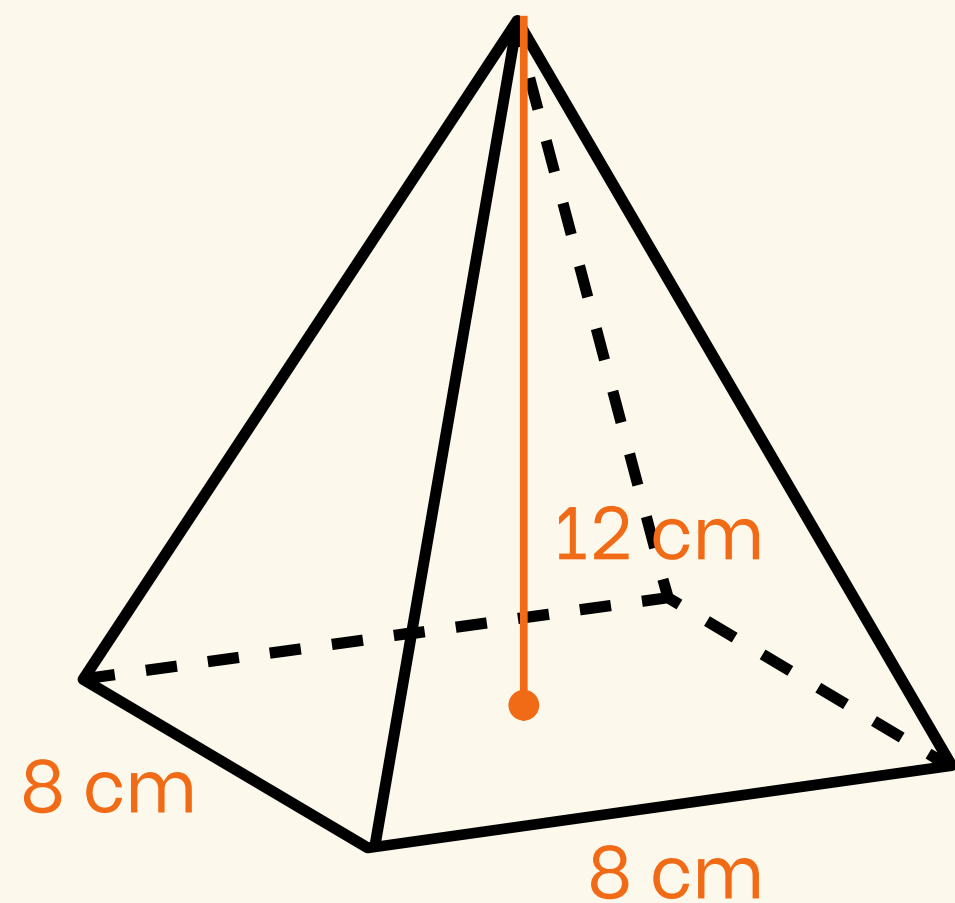


Sphere

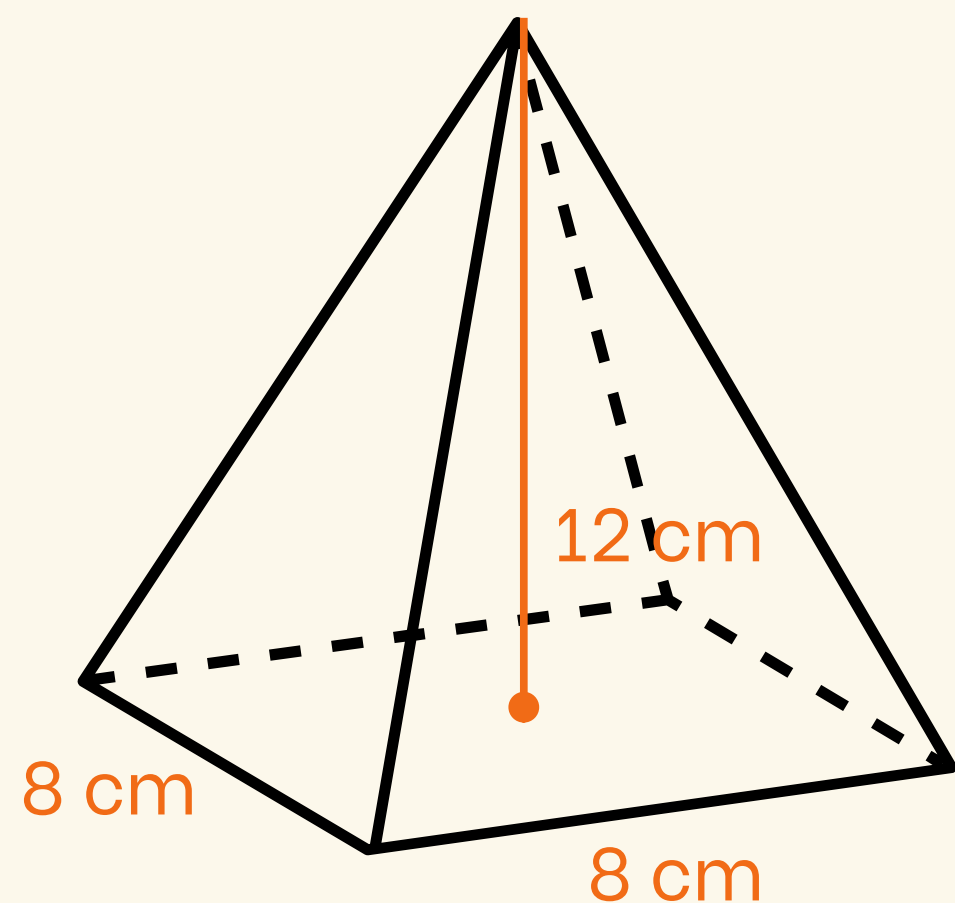
$$A = 201.06\text{in}^2$$



Classify each solid figure. Then, calculate the volume of each figure.

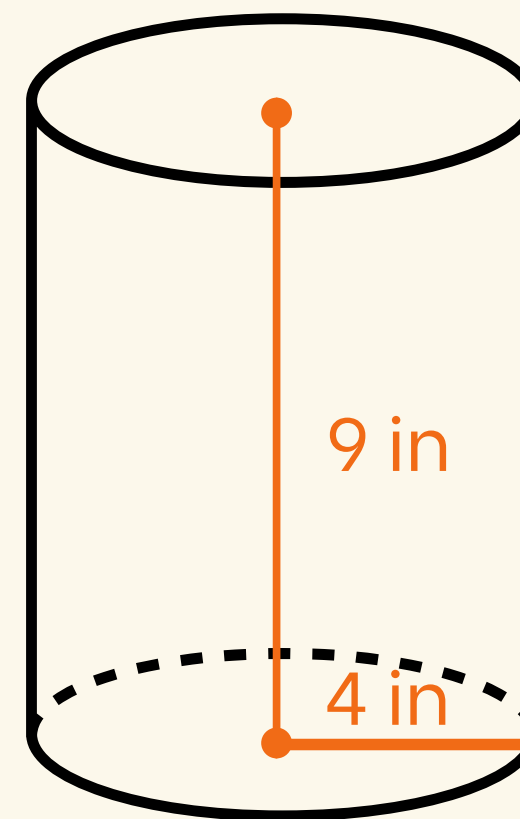


Classify each solid figure. Then, calculate the volume of each figure.



Pyramid

$$V = 256cm^3$$



Cylinder

$$V = 452.39cm^3$$



Key Points

Surface Area

- ✦ It is the total area of all the surfaces of a 3D figure.

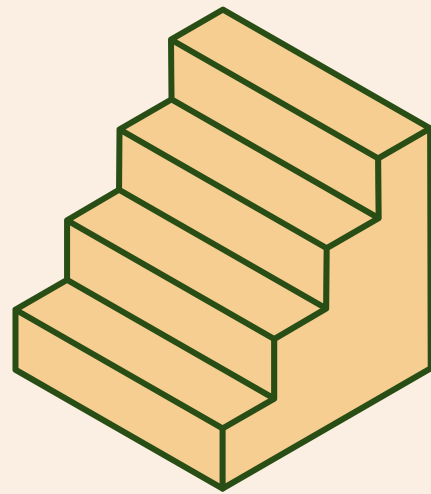
Volume

- ✦ It is the amount of space occupied by any 3D figure.

Formulas

- ✦ Each solid figure has different formulas to calculate their surface areas and volumes.

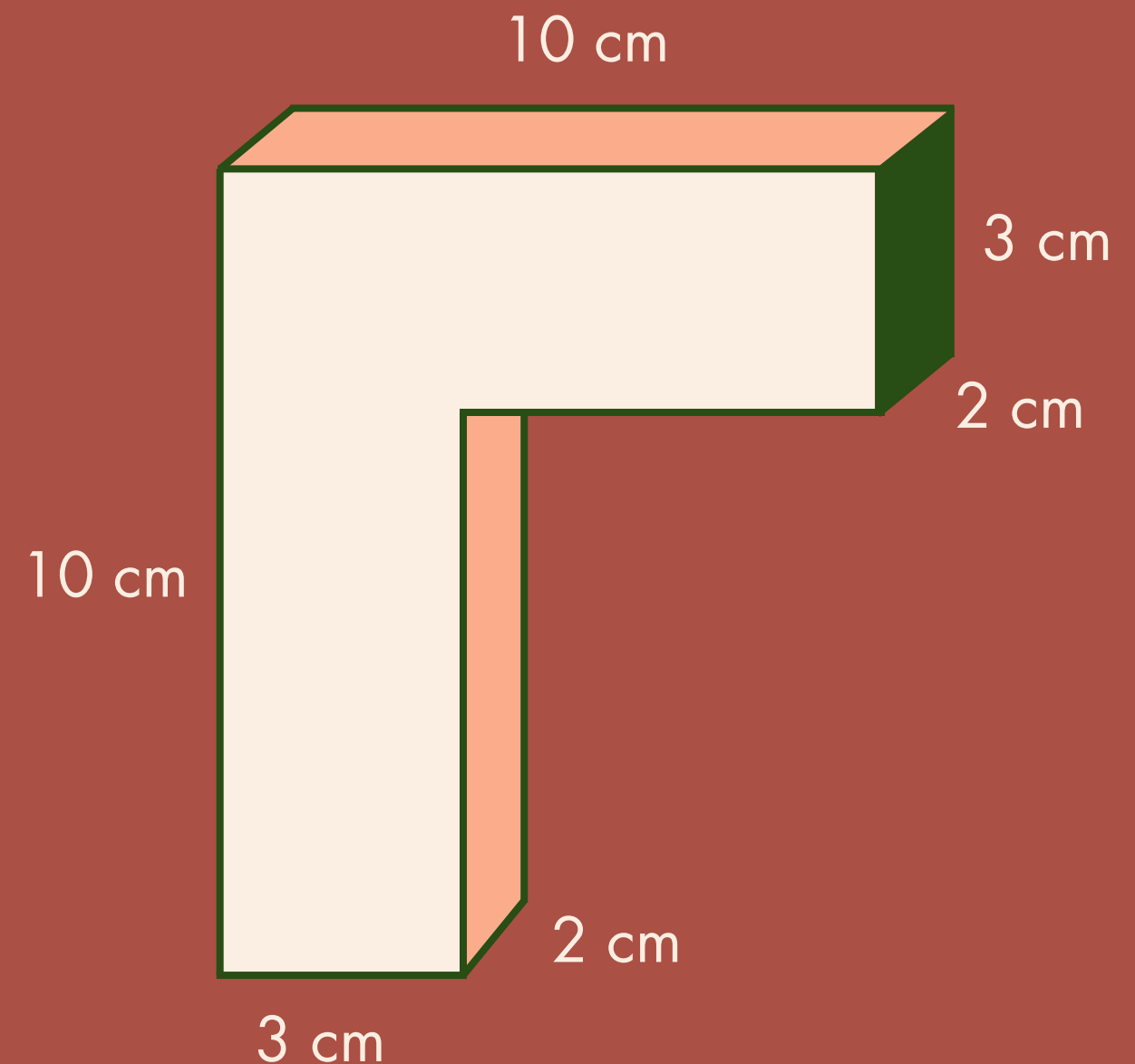
Volume of Composite Prisms



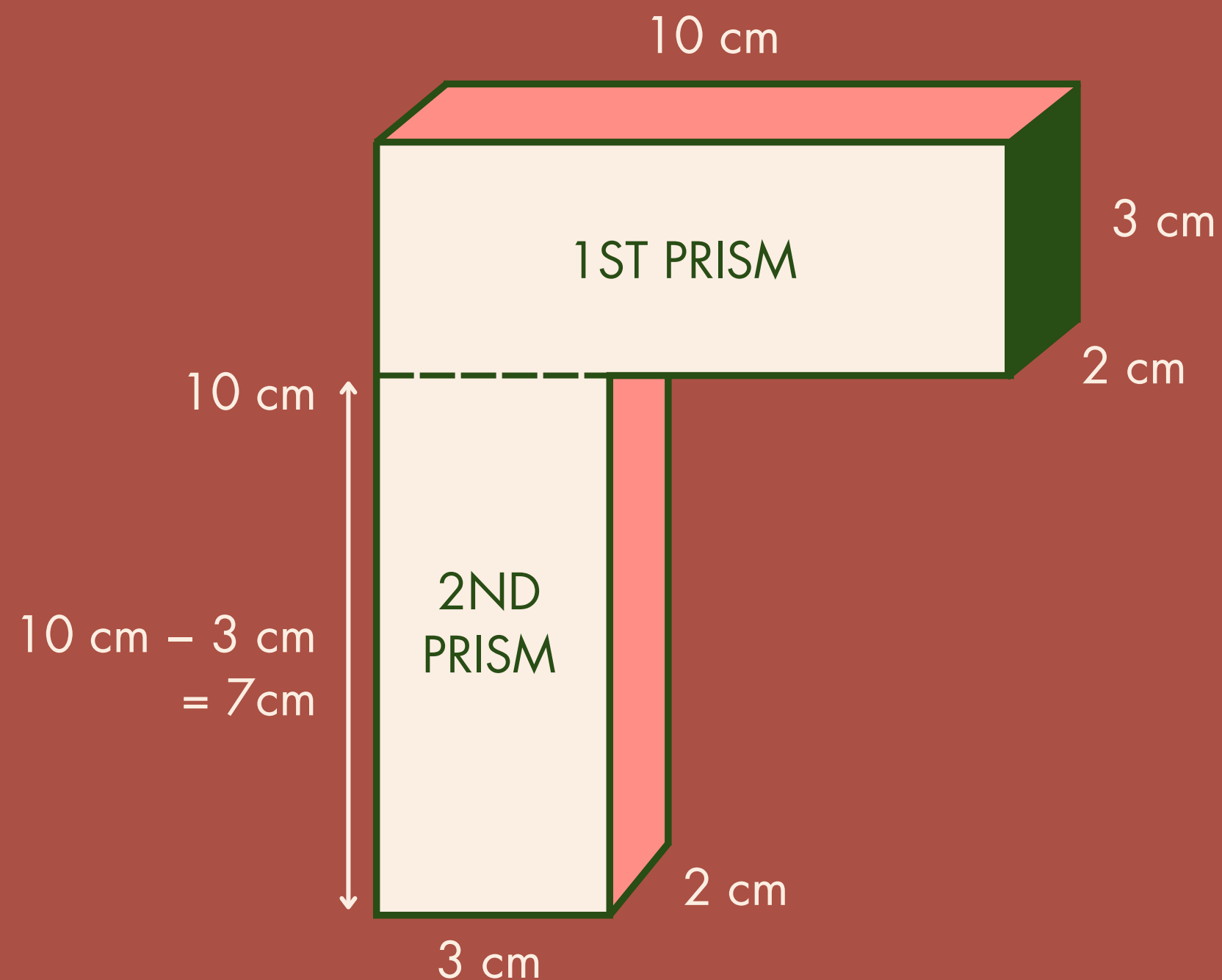
Composite Prism

- any prism made up of two or more prisms
- find its volume by adding all the volumes of rectangular prisms

What is the volume of the composite prism below?



What is the volume of the composite prism below?



Volume of 1st Prism:

$$V_1 = l \times w \times h$$

$$V_1 = 10 \text{ cm} \times 2 \text{ cm} \times 3 \text{ cm} = 60 \text{ cm}^3$$

Volume of 2nd Prism:

$$V_2 = l \times w \times h$$

$$V_2 = 7 \text{ cm} \times 2 \text{ cm} \times 3 \text{ cm} = 42 \text{ cm}^3$$

$$\text{Total Volume} = V_1 + V_2$$

$$\text{Total Volume} = 60 \text{ cm}^3 + 42 \text{ cm}^3$$

$$\text{Total Volume} = 102 \text{ cm}^3$$

Thank you!
You did great!

